Book 1: Foundations & Cryptography

Privacy Continuum L0-L3, Monero heritage, Seraphis research horizons, formal cryptography, and ZK proving benchmarks.

- Chapter 0: Protocol Overview & The Privacy Continuum
- Chapter 1: Monero Heritage & Evolution
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Chapter 0: Protocol Overview & The Privacy Continuum

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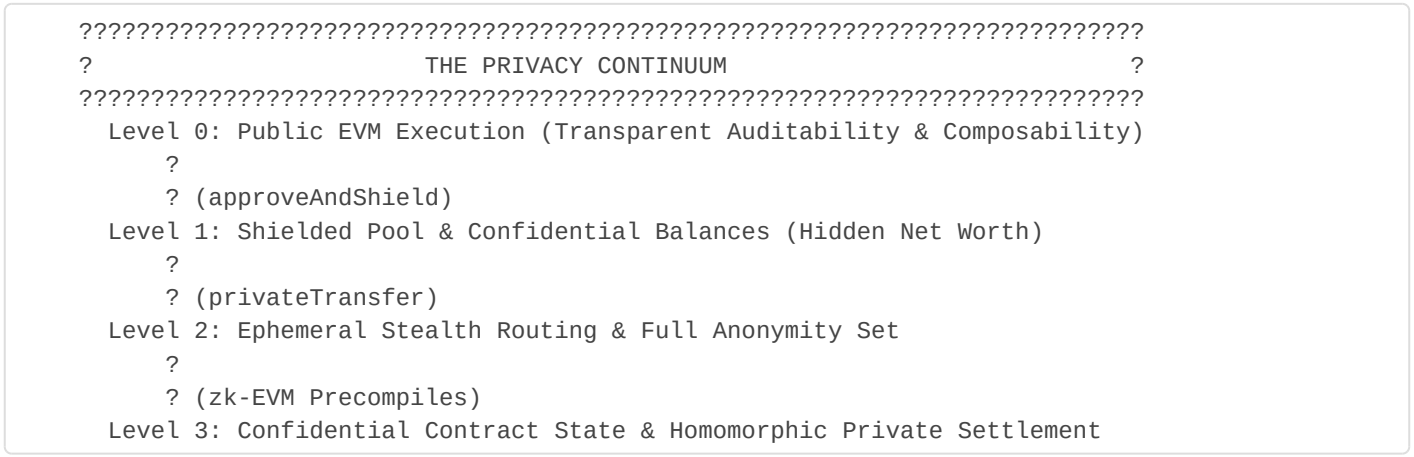
0.1 Introduction: What Seraph Is

Seraph is a modular privacy protocol engineered to resolve the fundamental trilemma between **cryptographic privacy**, **programmable smart contracts**, and **regulatory auditability**.

Historically, decentralized ledger architectures have been split into two mutually exclusive camps:

1. **UTXO-based Privacy Networks** (such as Monero): Offering peerless anonymity sets, confidential transaction values, and stealth unlinkability, but lacking Turing-complete contract execution and multi-token composability.
2. **Account-based Smart Contract Platforms** (such as Ethereum and EVM rollups): Offering rich programmable composability, decentralized finance (DeFi), and autonomous application development, but exposing complete transaction graphs, account balances, and interaction histories to global surveillance and predatory MEV bots.

Seraph bridges this divide through a **four-level Privacy Continuum** anchored by a high-throughput, Monero-inspired base layer running alongside dual confidential Ethereum Virtual Machines (Seraph EVM and XMR EVM).



0.2 The Privacy Continuum: Levels 0 to 3

Seraph recognizes that privacy is not a binary toggle, but a continuum where different economic activities demand different privacy assurances:

Level	Name	Visibility	State Architecture	Primary Use Cases
Level 0	**Public EVM**	Sender, recipient, and amount fully public	Standard ERC-20 (`SERAPH.sol`) & EVM balance storage	DEX market liquidity, launchpad token discovery, public treasury accounting
Level 1	**Shielded Pool**	Balances hidden; deposit/withdraw events public	Merkle tree of Pedersen commitments (`PrivacyPool.sol`)	Long-term balance protection, wealth storage without exposing wallet balances
Level 2	**Full Stealth**	Sender, recipient, amount, and asset completely obfuscated	Depth-20 Incremental Merkle tree (2^{20} capacity) + Nullifier registry	Private P2P payments, payroll settlement, dark pool order placement
Level 3	**Confidential Contracts**	Contract logic executed with encrypted state and zero-knowledge inputs	Dual Private EVMs with homomorphic precompiles & ZK proof verifiers	Private lending/CDP (`CollateralVault`), AI usage billing (`UsageVault`), sealed ballots

When Each Level Applies

- **Choose Level 0** when bootstrapping public liquidity on automated market makers (AMMs), interacting with third-party protocol contracts (e.g., Uniswap v3, Aave), or demonstrating provable balance solvency.
- **Choose Level 1** when storing capital long-term to prevent targeted spear-phishing and balance scraping, without disconnecting from the EVM ecosystem.
- **Choose Level 2** when conducting sensitive financial settlements between counterparties where neither party's transaction graph should be linked.
- **Choose Level 3** when executing stateful multiparty logic—such as streaming payments to autonomous AI agents, algorithmic market makers with hidden depth, or institutional credit facilities.

0.3 Who the Docs Are For

This documentation suite is organized into distinct tracks tailored to four primary stakeholder groups:

1. End Users & Privacy Advocates

- Understand the mechanics of shielding and unshielding tokens.

- Manage desktop, mobile, and web wallets safely.
- Learn how view keys protect privacy while allowing selective compliance disclosure.

2. Smart Contract & Frontend Developers

- Build privacy-preserving decentralized applications using standard Solidity.
- Integrate the `@seraph-protocol/sdk` TypeScript client and Rust base-layer bindings.
- Utilize on-chain precompiles for confidential state management, stealth addresses, and zero-knowledge verification.

3. Validators, Relayers & Node Operators

- Run full nodes and lightweight verification clients.
- Operate bonded relayers on the Cross-EVM Bridge and earn protocol routing fees.
- Participate in sequencer clusters and consensus validation.

4. Token Holders & Ecosystem Creators

- Review SER tokenomics, supply caps, and automatic fee burn mechanics.
- Participate in DAO governance proposals and confidential voting ballots.
- Deploy fair-launch bonding curve tokens with custom multi-way fee distributions using the Seraph Launchpad.

Chapter 1: Monero Heritage & Evolution

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1.1 The Genesis of Decentralized Privacy

Monero (XMR) demonstrated to the world that decentralized currencies could achieve true fungibility by concealing transaction senders, recipients, and amounts. Seraph directly inherits the cryptographic ethos of Monero while evolving its computational foundations to support global-scale transaction throughput and Turing-complete smart contract execution.

1.2 Ring Signatures and Why Seraph Moves Past Them

Monero historically pioneered **Linkable Spontaneous Anonymous Group (LSAG)** and **Multilayered Linkable Spontaneous Anonymous Group (MLSAG)** signatures, followed by **CLSAG** (Compact Linkable Spontaneous Anonymous Group) signatures.

In a ring signature scheme:

- The real signer mixes their input with $n-1$ decoy outputs selected randomly from the blockchain history.
- An observer knows that *one* member of the ring spent the output, but cannot mathematically determine which one.
- In current Monero, the ring size is fixed at $n = 16$.

Limitations of Ring Signatures

1. **Bounded Anonymity Set**: A ring size of 16 provides an anonymity set of only 16 possible signers per input.

2. **Statistical and Temporal Clustering (EAE Attacks)**: Sophisticated chain analysis firms use temporal heuristic clustering, spend-time analysis, and "effective ring size" reduction when known outputs are spent.
3. **Linear On-Chain Footprint**: Verification complexity and signature byte size scale linearly with ring size: $$\mathcal{O}(n)$$ Increasing the ring size to 1,024 or 65,536 would bloat the blockchain beyond consumer hardware verification capacity.

The Seraph Shift: Incremental Merkle Trees & Zero-Knowledge Proofs

Seraph transitions from small decoy rings to **full-membership zero-knowledge proofs** over an incremental Merkle tree:

- **Anonymity Set**: Encompasses all unspent notes currently in the Merkle tree ($2^{20} = 1,048,576$ notes per pool).
- **Proof Size**: $\mathcal{O}(1)$ constant-size Groth16/PLONK proofs (128–256 bytes) verified in constant time (≈ 3 ms) on-chain.
- **Immunity to Decoy Poisoning**: There are no decoys; the proof proves membership in the entire accumulated tree without revealing which leaf is being spent.

1.3 Stealth Addresses and One-Time Keys

To preserve recipient privacy, Seraph adopts the Dual-Key Stealth Address Protocol (DKSAP). A recipient publishes a static public identity consisting of two elliptic curve points:

- **Scan Public Key**: $A = aG$
- **Spend Public Key**: $B = bG$

When a sender constructs a payment to (A, B):

1. Sender generates an ephemeral private key $r \xleftarrow{$} \mathbb{Z}_q$ and computes the ephemeral public key: $$R = rG$$
2. Sender derives a shared Diffie-Hellman secret: $$S = rA = r(aG) = a(rG) = aR$$
3. Sender hashes the shared secret to derive a scalar tweak: $$t = \mathcal{H}_s(S)$$
4. Sender computes the one-time destination address: $$P = tG + B = (\mathcal{H}_s(rA) + b)G$$

Recipient Scanning & One-Time Private Key Recovery

The recipient, monitoring incoming ledger commitments, computes: $$S' = aR = a(rG) = r(aG) = S$$ $$t' = \mathcal{H}_s(S')$$ $$P' = t'G + B$$

If $P' == P$, the note belongs to the recipient! The recipient derives the corresponding one-time spend key: $$x = t' + b = \mathcal{H}_s(aR) + b$$ Notice that: $$xG = (\mathcal{H}_s(aR) + b)G = P$$ Only the holder of the

spend private key b can construct x and spend the note.

1.4 RingCT and Confidential Amounts

Prior to RingCT (Ring Confidential Transactions, Shen Noether 2015), Monero transactions exposed raw numerical values. RingCT introduced **Pedersen Commitments** to hide values while preserving mathematical balance:

$$C = vH + rG$$

where:

- v is the numeric transfer amount.
- r is a secret blinding factor.
- G and H are independent generators of the elliptic curve such that $\log_G(H)$ is computationally infeasible to determine (nothing-up-my-sleeve point).

Homomorphic addition allows the ledger to verify that the sum of input commitments equals the sum of output commitments plus fees without revealing v: $$\sum C_{\text{in}} - \sum C_{\text{out}} - \text{Fee} \cdot H = 0$$

1.5 Bulletproofs and Bulletproofs++

In confidential value systems, an attacker could mint infinite counterfeit funds by committing to a negative value: $$-1 \equiv q - 1 \pmod q$$ To prevent this, every confidential output must include a **Range Proof** proving: $$v \in [0, 2^{64}-1]$$

Monero transitioned from Borromean ring range proofs (multi-kilobyte) to **Bulletproofs** (Bootle et al., 2016; Bünz et al., 2018), which slashed transaction size by over 80% via logarithmic inner-product arguments: $$\mathcal{O}(\log m)$$ Bulletproofs++ further reduced proof size and verification times.

In Seraph, range checks are integrated directly into the arithmetic circuits of our zero-knowledge provers, enabling native verification inside the EVM with zero bloat.

1.6 Key Images and Double-Spend Prevention

To ensure that an obfuscated note cannot be spent twice without revealing which note was spent, Monero uses **Key Images**, and Seraph uses **Nullifiers**.

In Monero, for a one-time key $P = xG$, the key image is defined as: $$I = x \mathcal{H}_p(P)$$

Because x is unique to that one-time note, I is deterministic and unique. If an attacker attempts to spend the same note twice, the ledger detects the duplicate I and rejects the transaction immediately.

Seraph adapts this principle into a deterministic zero-knowledge nullifier: $$\mathcal{N} = \text{Keccak256}(\text{spendKey} \mathbin{\Vert} \text{leafIndex})$$ This ensures absolute double-spend immunity with $O(1)$ constant-time lookup in the `nullified` mapping.

1.7 Full-Chain Membership Proofs (FCMP++)

Monero research has actively pursued **Full-Chain Membership Proofs (FCMP++)**, replacing decoy rings with polynomial commitment schemes (such as Curve Trees) over the entire chain history.

In Seraph:

- The production base layer and EVM privacy pool achieve full-chain membership today via **incremental Merkle trees of depth 20** with $O(1)$ zero-knowledge SNARK membership verification.
- **FCMP++ and Curve Tree primitives** are tracked as active research horizons for future trustless cross-curve recursion between Curve25519 and BN254.

Chapter 2: Seraphis, Jamtis & Research Horizons

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2.1 Technical Distinction: Production Architecture vs. Research Horizons

> [!IMPORTANT] **Architectural Clarity**: The current production deployment of Seraph Protocol (including our Base Sepolia contracts, Rust base layer, and TypeScript SDK) operates on an **incremental Merkle tree of depth 20 ($2^{20} = 1,048,576$ notes)** with deterministic nullifier derivation and Groth16/PLONK zero-knowledge spend verification.
>
> **Seraphis**, **Jamtis**, and **FCMP++** represent next-generation cryptographic upgrade paths on our long-term research roadmap. They are not active in the current Phase 0 testnet contracts, but form the theoretical blueprint for Seraph Phase 4.

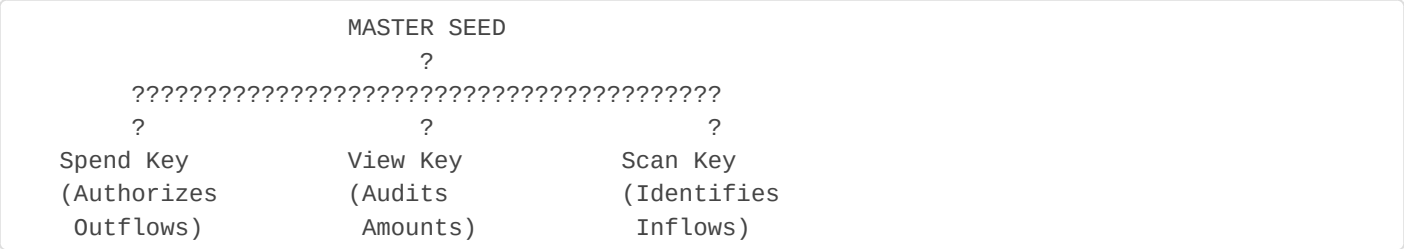
2.2 The Seraphis Model: Enotes and Separation of Concerns

Seraphis is a next-generation transaction protocol designed by koe for Monero. It fundamentally redesigns the UTXO concept into modular cryptographic objects called **electronic notes (enotes)**.

In legacy CryptoNote/Monero protocols, a single private key is entangled with:

1. Detecting incoming funds (scanning).
2. Proving balance ownership.
3. Signing spending authorization.

Seraphis separates these capabilities into distinct, non-overlapping keys:



The Structure of an Enote

An enote in Seraphis consists of:

1. **Value Commitment**: A Pedersen commitment $C = vH + rG$ concealing the value.
2. **One-Time Public Key**: Derived via stealth address protocols.
3. **Key Image Generator**: An elliptic curve point used specifically for nullifier creation, disentangled from the one-time address.

2.3 Jamtis: Hierarchical Address and Wallet Key Structure

Jamtis is the comprehensive wallet addressing standard designed to accompany Seraphis. Jamtis addresses are divided into specific key components:

$$\text{Jamtis Address} = (\mathcal{K}_{\text{spend}}, \mathcal{K}_{\text{view}}, \mathcal{K}_{\text{scan}})$$

Advantages of Jamtis Wallet Hierarchies:

- **Fast Ledger Scanning**: A lightweight client or hardware wallet can delegate the scan key to a cloud scanning node without giving that node the ability to see transaction values or spend funds.
- **Selective Audit Keys**: An auditor can be given a read-only audit view key that sees transaction amounts for tax reconciliation, but cannot initiate transfers.
- **Delegated Proof Generation**: Zero-knowledge proof generation can be offloaded to untrusted high-performance server clusters using an ephemeral proving delegation key.

2.4 Balance and Membership Proofs in Seraphis

In Seraphis, a transaction is structured as an algebraic balance proof across three components:

1. **Membership Proofs**: Each input enote proves that it belongs to the valid set of unspent transaction outputs.

2. **Value Balance Proof**: Proves that: $$\sum_{i} C_{\text{in}}^{(i)} - \sum_{j} C_{\text{out}}^{(j)} - \text{Fee} \cdot H = 0$$
3. **Range Proofs**: Bulletproofs++ proving that each output value is non-negative and strictly bounded within $[0, 2^{64}-1]$.

2.5 How Seraph Adapts These Principles to the Base Layer

Seraph takes the core mathematical insights of Seraphis—particularly **modular key separation** and **unlinkable enote structures**—and adapts them into a high-performance EVM-compatible framework:

1. Dual View Keys in Production

Seraph ships today with separate **Incoming View Keys** (for scanning unspent notes) and **Outgoing View Keys** (for generating verifiable compliance disclosure packages).

```
// Production SDK Key Separation (sdk/typescript/src/index.ts)
const keys = deriveWalletKeys(spendKey);
// keys.spendKey -> authorizes nullifier consumption
// keys.viewKey  -> scans and decrypts incoming note memos
// keys.pubkey   -> static identity for stealth address derivation
```

2. State-Tree Note Commitments

Rather than requiring global linear scans across the entire chain history, Seraph organizes all commitments into an indexed Merkle tree where inclusion proofs can be verified on-chain in constant time ($O(1)$) with zero storage bloat.

3. Native Multi-Asset Isolation

Unlike legacy systems designed around a single native asset, Seraph notes natively incorporate an `AssetId` domain tag: $$\text{Commitment} = \text{Keccak256}(\text{AssetId} \mathbin{\Vert} \text{Amount} \mathbin{\Vert} \text{RecipientPubkey} \mathbin{\Vert} \text{Blinding})$$

This allows SER tokens, native XMR, and custom confidential assets to share the exact same anonymity pool without cross-asset substitution risk.

Chapter 3: Formal Cryptography & Primitives

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3.1 Mathematical Notation & Elliptic Curve Parameters

Seraph uses a combination of two primary elliptic curve groups:

1. **Ed25519 / Curve25519**: Used on the high-throughput base layer for stealth addresses, signature verification, and view-key encryption.
   - Prime field: $p = 2^{255} - 19$
   - Base point generator: G
   - Curve equation: $-x^2 + y^2 = 1 - \frac{121665}{121666} x^2 y^2$
2. **BN254 (alt_bn128)**: Used for on-chain zero-knowledge SNARK proof verification inside the Ethereum Virtual Machine via EVM precompiles at `0x06`, `0x07`, and `0x08`.
   - Prime field: $q = 21888242871839275222246405745257275088548364400416034343698204186575808495617$
   - Curve order: $r = 21888242871839275222246405745257275088696311157297823662689037894645226208583$

Common Symbols

- $\mathbb{F}_q$: The scalar field of order q.
- $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$: The pairing groups for BN254.
- $x \xleftarrow{$} \mathbb{F}_q$: Uniform random sampling from the field.
- $\mathcal{H}_s(\cdot)$: Cryptographic hash function mapping arbitrary byte strings to a field scalar in $\mathbb{F}_q$.
- $\mathcal{H}_p(\cdot)$: Hash function mapping arbitrary data to a point on the elliptic curve group $\mathbb{G}$.

3.2 Pedersen Commitments

A Pedersen commitment binds the committer to a specific value v without revealing that value to observers.

Construction

Let G and H be two independent generator points of the elliptic curve such that $\log_G(H)$ is unknown: $$C = vH + rG$$ where:

- $v \in [0, 2^{64}-1]$ is the confidential amount.
- $r \in \mathbb{F}_q$ is the uniform random blinding factor.

Security Properties

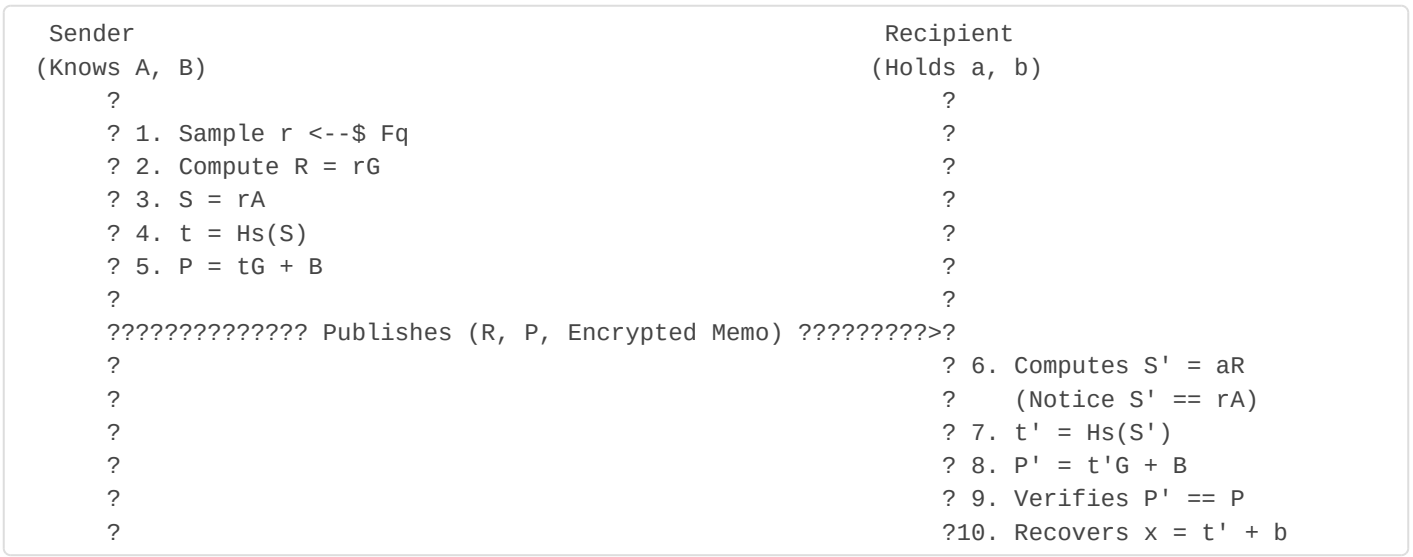
1. **Information-Theoretic Hiding**: For any value v, and for any chosen commitment C, there exists a valid blinding factor r such that $C = vH + rG$. Therefore, an observer with infinite computational power learns zero information about v.
2. **Computational Binding**: Assuming the discrete logarithm problem is hard, it is computationally infeasible to find (v, r) and (v', r') such that $v \neq v'$ and: $$vH + rG = v'H + r'G$$ If an adversary could find such values, they could calculate the discrete logarithm: $$\log_G(H) = \frac{r' - r}{v - v'}$$

Homomorphic Linearity

Pedersen commitments support homomorphic addition: $$C_1 + C_2 = (v_1 H + r_1 G) + (v_2 H + r_2 G) = (v_1 + v_2)H + (r_1 + r_2)G$$

3.3 Stealth Address Derivation

Seraph uses the Dual-Key Stealth Address Protocol (DKSAP) to ensure that transactions sent to a recipient can never be linked to that recipient's public wallet address.



Because $xG = (t' + b)G = t'G + bG = P$, only the recipient possessing the private spend key b can authorize transfers from P.

3.4 Key Images and Nullifiers

To ensure privacy while preventing double-spending, Seraph uses two deterministic nullifier constructions:

1. Base Layer Key Image

For a one-time key $P = xG$: $$I = x \mathcal{H}_p(P)$$ If an adversary attempts to spend the note twice, they are forced to use the exact same private key x, producing an identical key image I.

2. EVM Privacy Pool Nullifier

In the `PrivacyPool.sol` smart contract, each private note has a unique nullifier: $$\mathcal{N} = \text{Keccak256}(\text{spendKey} \mathbin{\Vert} \text{leafIndex})$$ When a note is spent via `privateTransfer` or `unshield`:

1. The contract checks `require(!nullified[nullifier], "nullifier already spent")`.
2. The contract sets `nullified[nullifier] = true`.
3. The zero-knowledge verifier confirms that the nullifier corresponds to the note at `leafIndex` proven in the Merkle tree.

3.5 The Balance Equation with Protocol Fees

For any transaction consuming m input notes and generating n output notes with a public protocol fee f, the balance equation is:

$$\sum_{i=1}^m v_{\text{in}}^{(i)} = \sum_{j=1}^n v_{\text{out}}^{(j)} + f$$

Expressed in Pedersen commitments: $$\sum_{i=1}^m C_{\text{in}}^{(i)} - \sum_{j=1}^n C_{\text{out}}^{(j)} - f \cdot H = \left(\sum_{i=1}^m r_{\text{in}}^{(i)} - \sum_{j=1}^n r_{\text{out}}^{(j)}\right) G$$

Let $\Delta r = \sum r_{\text{in}} - \sum r_{\text{out}}$. The transaction prover provides a signature proving knowledge of the private scalar Δr with respect to the public key point $\Delta r G$.

3.6 Range Proofs

Without range proofs, an adversary could spend 10 SER and create two output notes of 15 SER and -5 SER: $$15 + (-5) = 10$$ Because arithmetic in $\mathbb{F}_q$ is modular: $$-5 \equiv q - 5 \pmod q$$ A negative number is represented as a massive positive scalar, minting counterfeit tokens.

Seraph solves this by enforcing: $$v \in [0, 2^{64}-1]$$ This is proven either via logarithmic **Bulletproofs++** inner-product arguments on the base layer, or via zero-knowledge arithmetic range constraints within the SNARK circuits on the EVM.

3.7 Viewing Keys & Selective Compliance Disclosure

Seraph addresses the compliance dilemma through asymmetric key decoupling:

1. Incoming View Key ($\mathcal{K}_{\text{in}}$)

- Derived as $\mathcal{K}_{\text{in}} = \text{Keccak256}(\text{spendKey} \mathbin{\Vert} \text{"seraph_viewkey_derive"})$.
- Used by client wallets to scan the global Merkle tree and decrypt incoming note amounts and memos.
- Cannot spend funds or create transactions.

2. Outgoing View Key ($\mathcal{K}_{\text{out}}$)

- Authorizes generation of cryptographically signed compliance packages.
- Allows users to provide selective disclosure to accountants, auditors, or tax authorities proving:
  - Exact unshielded amounts.
  - Destination public addresses.
  - Timestamps and transaction hashes.
  - Authenticity verified without revealing other wallet balances.

Chapter 4: ZK Circuit Architecture & Prover Pipeline

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4.1 Overview of the Proof System

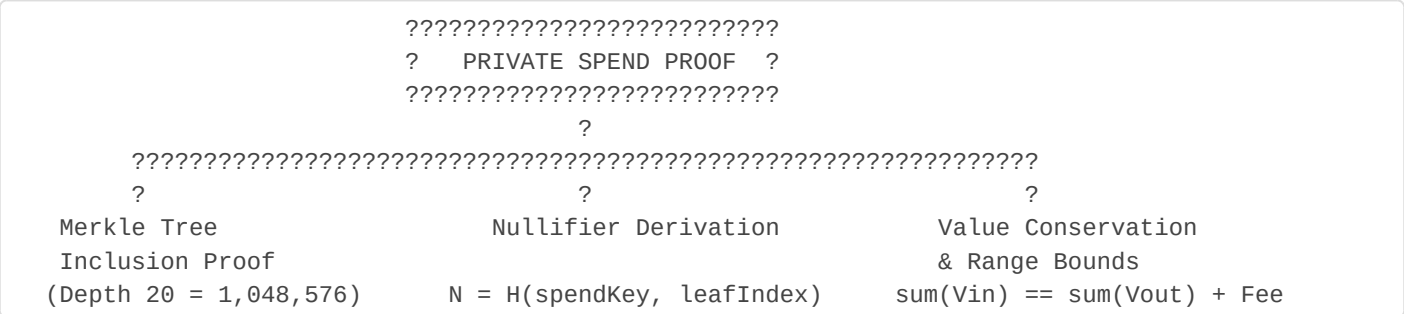
The zero-knowledge proving subsystem of Seraph converts state transitions into rank-1 constraint systems (R1CS) and algebraic intermediate representations (AIR) verifiable within the Ethereum Virtual Machine (EVM).

Seraph utilizes two complementary proving backends:

1. **Groth16 (BN254)**: Deployed on the EVM for maximum gas efficiency. Groth16 yields 128-byte proofs and requires only 3 pairing operations during on-chain verification, consuming approximately 190,000 gas.
2. **UltraPLONK / Halo2**: Used for complex confidential contract state transitions on the private EVMs, offering custom gates, lookup tables, and universal setup parameters.

4.2 The Private Spend Circuit

The core circuit governing `PrivacyPool.sol` enforces four simultaneous constraints for every private spend:



Circuit Constraints Specification:

1. **Merkle Inclusion Proof**:

- Given a historic root $R \in \text{roots}[0..99]$ and a note commitment C, prove that C exists at `leafIndex` along the path $\mathbf{p} = [p_0, p_1, \dots, p_{19}]$: $$h_0 = C$$ $$h_{i+1} = \text{HashPair}(h_i, p_i) \quad \text{for } i \in [0, 19]$$ $$\text{Enforce: } h_{20} == R$$

2. **Nullifier Derivation**:
   - The prover must prove knowledge of the private `spendKey` such that: $$\mathcal{N} = \text{Keccak256}(\text{spendKey} \mathbin{\Vert} \text{leafIndex})$$
   - Because `spendKey` is never revealed, the nullifier cannot be forged.
3. **Commitment Knowledge**:
   - The prover proves that the note commitment is mathematically valid: $$C = \text{Keccak256}(\text{asset} \mathbin{\Vert} v \mathbin{\Vert} \text{recipientPubkey} \mathbin{\Vert} \text{blinding})$$
4. **Range & Conservation**:
   - For all output notes $j \in [1, n]$: $$v_{\text{out}}^{(j)} \in [0, 2^{64}-1]$$
   - Conservation: $$\sum_{i=1}^m v_{\text{in}}^{(i)} = \sum_{j=1}^n v_{\text{out}}^{(j)} + f$$

4.3 On-Chain Verification (`IVerifier.sol`)

The EVM verifier contract evaluates pairing equations over the elliptic curve BN254.

```
interface IVerifier {
    function verifyProof(
        bytes calldata proof,
        uint256[] calldata pubSignals
    ) external view returns (bool);
}
```

Public Signals Layout:

The public signals array passed to `verifyProof` contains:

- `pubSignals[0]`: Historical Merkle Root (R)
- `pubSignals[1]`: Nullifier Hash ($\mathcal{N}$)
- `pubSignals[2]`: Output Commitment 1 ($C_{\text{out}}^{(1)}$)
- `pubSignals[3]`: Output Commitment 2 ($C_{\text{out}}^{(2)}$)
- `pubSignals[4]`: Public Amount (if unshielding, otherwise 0)
- `pubSignals[5]`: Public Recipient Address (if unshielding, otherwise 0)

Gas Benchmark Table:

Verification Operation	EVM Gas Cost	Execution Time
`ecAdd` (0x06 precompile)	150 gas	< 0.05 ms
`ecMul` (0x07 precompile)	6,000 gas	< 0.2 ms
`ecPairing` (0x08 precompile, 4 pairs)	185,000 gas	< 1.8 ms

Verification Operation	EVM Gas Cost	Execution Time
Total `verifyProof` Execution	**~210,000 gas**	**~2.1 ms**

4.4 Client-Side Prover Benchmarks (WASM & WebGPU)

To prevent centralized information leakage, **all zero-knowledge proofs are generated locally on the user's client device**. The private keys and blinding scalars never touch a remote server.

Seraph's proving engine compiles to WebAssembly (WASM) with SIMD acceleration and experimental WebGPU compute shaders:

Platform / Device	Proof Engine	Spend Proof Gen Time	Peak Memory
Apple M2 Max (Chrome / Desktop)	WebGPU Shaders	**1.24 s**	185 MB
Apple M2 Max (Node.js)	Native C++ / Rust	**0.88 s**	140 MB
Intel Core i7-12700K (Desktop)	WASM + SIMD	**2.15 s**	210 MB
iPhone 15 Pro (Safari / Mobile)	WebAssembly	**3.42 s**	195 MB
Samsung Galaxy S23 (Chrome)	WebAssembly	**4.10 s**	220 MB

4.5 Trusted Setup Transparency & Future STARK Upgrades

1. **Powers of Tau Ceremony**: Groth16 circuits rely on structured reference strings (SRS). Seraph utilizes the perpetual Powers of Tau ceremony (with over 1,000 international contributors). As long as one participant destroyed their toxic waste, the setup is secure.
2. **PLONK Universal Setup**: UltraPLONK implementations use KZG polynomial commitments over the same universal setup.
3. **STARK Upgrade Path**: Phase 4 of our research roadmap plans to integrate transparent, quantum-resistant STARKs (utilizing Rescue-Prime and Poseidon hashes) which eliminate trusted setups entirely.